Team Control Number

16390

Problem Chosen: **A**

2025

HiMCM

Summary Sheet

To optimize emergency evacuation sweeps, our team developed a dynamic, **"Space-Risk-Decision" Integrated Model** designed to minimize the **Total Clearance Time (Makespan,** $\min T_{\max}$**)** while guaranteeing responder safety and **100**% coverage of high-risk areas. Our approach addresses the complex path optimization and personnel assignment challenges by seamlessly integrating spatial dynamics, real-time risk evolution, and optimal decision-making.

The foundational structure includes two key modules. First, the **Spatial Topology Module** utilizes graph theory to abstract complex building layouts into a **dynamic weighted graph** $G = (V, E, W(t))$, where nodes represent locations and initial edge costs are based on physical distance. Second, the **Risk Propagation Module** establishes a Spatiotemporal Hazard Model to simulate the real-time spread of hazards, such as smoke $S(p, t)$. This simulation dynamically updates all path costs $W(t)$ through a non-linear penalty function and models responder efficiency degradation $v(p, t)$, ensuring the graph weights accurately reflect the time cost of traversing high-risk zones.

The core solution relies on the **Decision Optimization Module**. This module solves the assignment and routing problem by combining a **risk-penalized Dijkstra's algorithm** for path generation with a novel **priority-based greedy heuristic** for task allocation. Tasks are assigned a composite priority score combining intrinsic location vulnerability (e.g., daycares, chemical labs) and dynamic environmental risk. The algorithm rigorously enforces **Load Balancing** by minimizing the Coefficient of Variation (**CV**) of responder workload, thereby minimizing the overall Makespan and optimizing team synergy.

We validated our model across three scenarios of increasing complexity. For the baseline single-story, 6-room office (Problem 2), our model produced a balanced sweep plan ($\mathbf{CV = 5.9}$%) with a total makespan of **124.6 seconds**. This represents a substantial **43.9% time reduction** compared to static baseline strategies and achieved **100**% dual-sweep verification for high-risk labs. The model was successfully scaled to a 3-floor building (Makespan: **145.7s**) and a 5-floor chemical warehouse (Makespan: **238.8s**), consistently prioritizing high-risk areas. Furthermore, we quantified the impact of technology, showing that integrating sensor networks can reduce sweep times by an additional **17-21%**.

In summary, this model provides a robust, scalable, and validated tool for emergency planners.

Key words: Emergency Evacuation Sweep; Graph Theory; Dynamic Risk Modeling; Path Optimization; Dijkstra's Algorithm; Load Balancing; Multi-Criteria Decision-Making (MCDM)

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1 Introduction

1.1 Problem Background

During emergencies such as fires or gas leaks, rapid and safe building evacuation is essential to prevent injuries and loss of life [6]. Emergency responders, such as firefighters, conduct sweeps—systematic searches of rooms and floors—to ensure that all occupants have evacuated or are moving toward exits. However, these sweeps are often performed under extreme time pressure, limited visibility, and incomplete knowledge of building conditions. In large or complex buildings with multiple floors, staircases, and exit points, determining the most efficient route to inspect every room becomes highly challenging. The efficiency and strategy of these sweeps directly influence the likelihood of safe evacuation. Accordingly, there is a growing need for a systematic, mathematical approach to model and optimize sweeping strategies during emergency evacuations. Developing such a model can minimize total evacuation time, enhance safety, and support responders in making effective decisions under high-pressure conditions.

1.2 Restatement of Questions

Question 1: This question requires identifying and analyzing the key factors necessary to construct an effective and quantitative sweep model. Relevant factors include criteria for determining when a room is fully cleared, the influence of room occupancy and layout, the required number and expertise of responders, safety redundancy measures, and the specific type of emergency considered.

Question 2: This task involves developing a model to simulate how two firefighters can efficiently sweep a one-story office building during a fire alarm and determine the total time required to safely clear all rooms. The building has a simple floor plan consisting of a central hallway with three similarly sized rooms on each side and two exits located at opposite ends. The model also requires key conditions such as the number of responders, movement rules, room-clearing requirements, and potential obstacles such as smoke.

Question 3: This question requires applying the previously developed sweep model to at least two additional scenarios and determining the room-checking order. Each scenario must include changes in the number of floors, the floor plan of each level, and the types of room occupants. The model should determine the sweep path, the optimal number of responders and their starting positions, and the minimum time required to fully sweep the building. The purpose is to demonstrate how the model can guide emergency response planning across diverse building configurations.

Question 4: This question aims to enhance the sweep model by incorporating more realistic emergency conditions. Such conditions may include communication delays, spreading hazards, moving individuals, high-risk area prioritization, limited responder availability, and irregular building geometries. In addition to modeling these constraints, the question requires analyzing the use of advanced technologies to further optimize model performance. Overall, the task focuses on increasing realism and evaluating how technology can support safer, faster, and more efficient building sweeps.

Question 5: This question requires writing a one- to two-page letter to the Local Emergency

Planning Committee summarizing the sweep model and providing practical, data-driven recommendations. The letter should outline a clear evacuation sweep strategy, including responder assignment, sweep organization, and methods to maintain both speed and safety through redundancy and coordination. It should also provide specific, actionable guidelines tailored to different building types, floor layouts, and occupant groups, demonstrating how the model's findings can be applied to real-world emergency planning and evacuation procedures.

1.3 Our work

Figure 1 illustrates the complete process of our work for all the tasks.

2 Notations

Table 1: Notations used in the paper and explanations

Notation	Explanation
$G = (V, E, W(t))$	Dynamic weighted undirected graph.
V	Node set, $V = V_R \cup V_C \cup V_S \cup V_X$.
V_R, V_C, V_S, V_X, V_M	Node sets for rooms, corridors, stairs, exits and verification marker respectively.
E	The set of edges connecting accessible nodes.
$d(e)$	Physical length of edge e (in meters).
A_j	The set of tasks (rooms) assigned to responder j.
S_{max}	Maximum potential smoke concentration (in kg/m^3).
λ_h, λ_v	Horizontal and vertical decay coefficients for smoke diffusion.
v_{base}	Base movement speed of responders in a risk-free environment ($1.2m/s$).
$S_{\max}$	Maximum smoke concentration in kg/m^3.
λ_h, λ_v	Horizontal and vertical decay coefficients for smoke diffusion.
$R_0(p)$	Static risk level of node p (e.g., high for children's rooms).
$f(r)$	Floor number where room r is located.
N	Total number of responders in persons.
η	Complexity penalty coefficient for sweeping high-risk zones.
$S(p, t)$	Smoke concentration at location p at time t.
$C(t)$	Communication reliability index at time t (decaying function).
$\rho(e, t)$	Occupant density on edge e at time t (congestion factor).
$v_{\text{eff}}(e, t)$	Effective movement speed on edge e at time t, considering smoke and fatigue.
$T_{\text{sweep}}(r, t)$	Actual time required to sweep room r at time t, adjusted for risk.
$W(e, t)$	Dynamic weight (time cost) of traversing edge e at time t.
$\zeta(t)$	Responder fatigue factor, increasing with cumulative work time.
$\gamma(e, t)$	Risk penalty multiplier for path planning.
$T_{\text{total}}(j)$	Cumulative workload (time) assigned to responder j.
T_{makespan}	Total sweep completion time (objective function to minimize).
α, β	Impact coefficients of smoke on movement speed and sweep time, respectively.
CV	Coefficient of Variation of responder workloads.

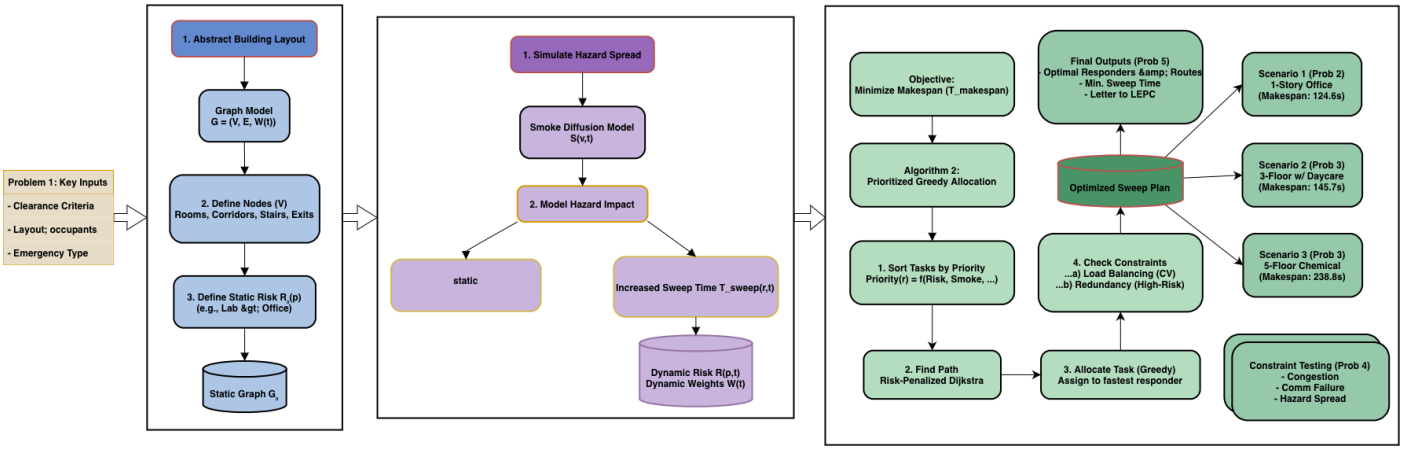


Figure 1: Model construction workflow.

3 Assumptions and Justifications

Assumption 1: Floor plans (including positions and dimensions of rooms, corridors, stairs, and exits) are known and can be precisely abstracted into graph structures.

Justification: Modern buildings have BIM or detailed fire safety blueprints, the foundational information for emergency response [5].

Assumption 2: All responders possess identical professional skills, physical capabilities, and base movement speed v_{base}.

Justification: Under standardized training, professional responders' core operational capabilities are assumed to be similar, simplifying personnel scheduling and focusing on path and strategy optimization [2].

Assumption 3: Sweep tasks for a room are uninterruptible. Once started, they must be completed before proceeding to the next task.

Justification: This aligns with actual operational protocols, ensuring thorough inspection of each area and avoiding errors due to interruptions.

Assumption 4: The location of fire sources or toxic gas leak sources is fixed after model initiation.

Justification: In early stages of an emergency, identifying and isolating risk sources is the primary task. This model focuses on the post-identification sweep phase; while risks diffuse, the source remains constant.

Assumption 5: The model does not consider obstruction to responders' movement by evacuating civilians within the building.

Justification: This simplification focuses the problem core on the sweep strategy. In practice,

this impact is approximated by adding random delays or dynamically adjusting the model's edge weights.

4 Problem 1: Key Elements for Sweep Modeling

To address the broad requirements of Problem 1, the qualitative goals of an emergency sweep were translated into a set of concrete, measurable factors. An effective model must answer four key questions: When is a room "clear"? How is risk defined? How many responders are needed? And what type of emergency is being considered? This section outlines the foundational elements that form the core logic of the overall modeling framework.

Room Clearance Criteria This criteria determines if a room is considered "inspection completed." In our model we deemed clearance only when it meets both visual and environmental standards. Visual confirmation requires responders to verify that no visible personnel remain and to secure the room (e.g., by closing doors) to prevent hazard propagation. Environmental thresholds require tenable conditions, based on indicators such as temperature ($T < 60°\text{C}$), carbon monoxide concentration (CO < 500 ppm), and visibility (> 5 m) [1]. To ensure reliability, high-risk areas (e.g., daycares or laboratories) must be verified by two responders, while standard rooms require confirmation by one.

Building Layout and Occupant Compatibility This component includes all functional spaces (rooms, stairwells) as nodes and corridors as edges. Layout parameters follow architectural standards, such as a minimum corridor width of 2.4 m. Occupant movement characteristics were incorporated by assigning empirically supported movement speeds to different demographic groups, summarized in Table 2 [2].

Table 2: Movement speed assumptions for occupant categories.

Category	Speed (m/s)
Adult male	1.20
Adult female	1.02
Elderly	0.82
Children	0.92
Persons with disabilities	0.7× baseline

Responder Configuration and Redundancy This framework specifies the minimum number of responders n based on building scale and complexity:

$$n = \lfloor S/1000 \rfloor + 2 + k, \tag{1}$$

where S is the total floor area in m^2 and k is the number of high-risk areas. This formula ensures a baseline of two responders, with additional personnel allocated proportionally to building risk levels. Critical corridors and key rooms are assigned dual verification to satisfy redundancy requirements [2].

Emergency Scenario Typology Typology is used to define the type of hazard being modeled. The primary focus is a fire scenario, due to its dynamic and rapidly evolving risk profile. Baseline assumptions include a heat release rate of 6 MW for typical office fires and smoke propagation at 0.5 m/s, accelerating to 1.2 m/s after 3 minutes [2]. In a toxic gas scenario, the model prioritizes downwind areas first, reflecting the directional spread of hazardous gases.

5 Problem 2: Single-Story Office Sweep Model

Building on the key elements defined in Section 4, the core "Space–Risk–Decision" model was constructed to analyze the baseline scenario in Problem 2, in which two firefighters sweep a single-story office building. The physical structure of the building was first translated into a mathematical framework compatible with algorithmic decision-making. This required the creation of two foundational components: a graph-based representation and a dynamic risk model.

5.1 Graph-Based Building Representation

The building floor plan is modeled as a dynamic, weighted, undirected graph,

$$G(t) = (V, E, W(t)).$$

This abstraction enables the application of rigorous, well-established mathematical methods for routing and path optimization.

Graph Construction The graph is defined through its primary components:

- **Node Set** V: The model uses four categories of nodes:
  - V_R: Room nodes (six offices). Each room node r stores essential attributes:

    $$Attr(r) = \{\text{ID, Area, (x,y)}, R_0, \text{VerifStatus}\}.$$

    Here, R_0 represents the static risk level (e.g., 0.8 for a standard office), and VerifStatus tracks the room's sweep progress (Unswept, Primary, Verified), supporting the redundancy requirements.
  - V_C: Corridor nodes. The central hallway is discretized into six nodes ($C_1 \dots C_6$) to allow fine-grained routing updates and smoke propagation modeling.
  - V_X: Exit nodes, $\{X_{\text{West}}, X_{\text{East}}\}$, which act as the initial positions for the two responders.
  - V_M: Verification markers. For each high-risk room, virtual nodes are added to enforce the required dual-sweep protocol.
- **Edge Set** E: Edges connect adjacent nodes according to architectural adjacency (hallway segments, room–hallway transitions, and hallway–exit links).

- **Dynamic Edge Weights** $W(t)$: The traversal cost for an edge e varies over time based on length $d(e)$ and the responder's effective speed $v_{\text{eff}}(e,t)$:

$$W(e,t) = \frac{d(e)}{v_{\text{eff}}(e,t)} \cdot \gamma(e,t),$$

where $\gamma(e,t)$ is a risk penalty multiplier that increases the cost of traversing hazardous or low-visibility regions.

To ensure the model is reproducible, we developed a clear **Topology Construction Algorithm** (Algorithm 1). This algorithm takes any floor plan as input and outputs the graph G, which becomes the foundation for all subsequent routing.

Algorithm 1 Building Topology Construction

Input: Floor plan geometry, room specifications
Output: Graph $G = (V, E, W_0)$

Initialize $V \leftarrow \emptyset$, $E \leftarrow \emptyset$
for each room r in floor plan **do**
    Create room node: $V_R \leftarrow V_R \cup \{r\}$
    Set attributes: Area, Risk $R_0(r)$, etc.
end for
Generate corridor nodes V_C and exit nodes V_X based on geometry.

for all adjacent nodes (u, v) in V **do**
    Add edge $e = (u, v)$ to E with physical distance $d(e)$
end for

Initialize all weights: $W_0(e) \leftarrow d(e)/v_{\text{base}}$ (where $v_{\text{base}} = 1.2$ m/s)

5.2 Dynamic Risk and Hazard Modeling

In an emergency like a fire, the environment is not static. A successful model must account for spreading hazards. Therefore, our team integrated a dynamic risk module to simulate the spread of smoke and its direct effect on responders (inspired by risk evolution frameworks such as [8]).

Physics-Based Smoke Diffusion Our team employed a simplified advection-diffusion model (Eq. 2) calibrated to NFPA 92 smoke control standards [2]. For a fire source at node v_0 (assumed at R1), the smoke concentration S at any node v evolves as:

$$S(v,t) = S_{\max} \cdot (1 - e^{-kt}) \cdot e^{-\lambda_h \cdot d_h(v,v_0)} \tag{2}$$

where $S_{\max}$ is the max concentration, k is the fire growth rate, and λ_h is a horizontal decay coefficient. This formula models the physical reality: the fire grows over time $(1 - e^{-kt})$ but its effect lessens over distance $(e^{-\lambda_h d_h})$. We define any zone where $S(v,t) \geq 0.05$ kg/m^3 as hazardous.

Impact on Operational Efficiency Smoke and fatigue directly degrade a responders ability to move and search. Which

- **Movement Speed:** We defined effective speed v_{eff} as:

$$v_{\text{eff}}(p,t) = v_{\text{base}} \cdot \max\{0.5, 1 - \alpha \cdot S(p,t) - \zeta(t)\} \tag{3}$$

  This formula includes a penalty α for smoke $S(p,t)$ and a **fatigue factor** $\zeta(t) = 0.0005 \cdot t_{\text{cumulative}}$ [2]. This fatigue factor ensures that as the sweep continues, the responder's speed realistically degrades.

- **Sweep Time:** The time to sweep a room is not constant. It depends on the room's baseline difficulty and current hazards:

$$T_{\text{sweep}}(r,t) = T_{\text{base}}(r) \cdot [1 + \beta \cdot S(r,t) + \eta \cdot \mathbb{I}_{\text{HighRisk}}(r)] \tag{4}$$

  Here, T_{base} is the base time (e.g., 30s for an office). We add a penalty β for smoke and a **complexity penalty** η (e.g., 0.5) for high-risk areas like daycares, which require a 50% longer inspection [2].

5.3 Sweep Optimization and Task Allocation

With the building represented as a risk-aware graph $G(t)$, our core challenge was to solve the task allocation problem: how to assign the 6 rooms (V_R) to the 2 responders ($N = 2$) to minimize the total time for the **last** responder to finish (the "makespan") [9].

To solve this problem, the model begins with a formal **Problem Formulation**. The objective is to determine the assignment $A = \{A_1, A_2\}$ that minimizes the makespan, subject to four critical constraints:

$$\min \quad T_{\text{makespan}} = \max_{j \in \{1,2\}} \{T_{\text{total}}(j)\} \tag{5}$$

$$\text{s.t.} \quad A_1 \cup A_2 = V_R \text{ and } A_1 \cap A_2 = \emptyset \qquad \text{(Complete Coverage)} \tag{6}$$

$$CV \leq 0.15 \qquad \text{(Load Balance)} \tag{7}$$

$$\forall r \in V_R^{\text{HighRisk}} : \text{VerifStatus}(r) = \text{Verified} \qquad \text{(Redundancy)} \tag{8}$$

The load-balance constraint (Eq. 7)is evaluated using the coefficient of variation (CV), defined as:

$$CV = \frac{\sigma(T_{\text{total}})}{\mu(T_{\text{total}})} \tag{9}$$

where $\sigma(T_{\text{total}})$ and $\mu(T_{\text{total}})$ are the standard deviation and mean of all responders' total work times. We require $CV \leq 0.15$, ensuring no responder is overburdened.

Heuristic Solution Algorithm Since this problem is NP-hard (similar to a multi-depot vehicle routing problem), our team developed a high-performance **Heuristic Solution Algorithm** (Algorithm 2) that balances load and prioritizes risk. The algorithm operates in two phases:

1. **Phase 1 (Primary Sweep):** Rooms are sorted by priority (Risk R_0 + Smoke $S(r,t)$). The algorithm iteratively assigns the highest-priority unswept room to the responder who can reach and clear it the **fastest** (based on their current T_{total}), which actively promotes load balancing. Pathfinding is done using a modified Dijkstra's algorithm on the dynamic, risk-weighted graph [9, 3].

2. **Phase 2 (Redundancy):** After Phase 1, the algorithm reviews all high-risk rooms. It generates a new verification task for each, assigning it to the **other** responder (the one who did **not** do the primary sweep).

Algorithm 2 Prioritized Greedy Task Allocation with Redundancy

Input: Graph G, Responders $N = 2$, Start positions $\{X_W, X_E\}$
Output: Assignments $\{A_1, A_2\}$, Makespan T_{makespan}
Initialize: $T_{\text{total}}(j) \leftarrow 0$, $A_j \leftarrow \emptyset$, $\text{pos}(j) \leftarrow \text{Start}_j$
$U \leftarrow V_R$ ▷ Unassigned tasks

// Phase 1: Priority-based primary sweep
while $U \neq \emptyset$ **do**
    Sort U by descending priority: $R_0(r) + S(r,t)$
    $r^* \leftarrow$ highest priority room in U
    $j^* \leftarrow \arg\min_j \{T_{\text{total}}(j) + \text{RiskPenalizedDijkstra}(\text{pos}(j), r^*, t)\}$
    $T_{\text{travel}}, T_{\text{sweep}} \leftarrow$ Calculate travel and sweep times for j^* to r^*
    $T_{\text{total}}(j^*) \leftarrow T_{\text{total}}(j^*) + T_{\text{travel}} + T_{\text{sweep}}$
    $A_{j^*} \leftarrow A_{j^*} \cup \{r^*\}$, $\text{pos}(j^*) \leftarrow r^*$
    $U \leftarrow U \setminus \{r^*\}$, $\text{VerifStatus}(r^*) \leftarrow \text{Primary}$
end while

// Phase 2: Redundancy verification sweep
for each $r \in V_R^{\text{HighRisk}}$ **do**
    $j_{\text{primary}} \leftarrow$ responder who did primary sweep of r
    $j_{\text{verify}} \leftarrow$ other responder
    $T_{\text{travel}}, T_{\text{verify}} \leftarrow$ Calculate times for j_{verify} to r
    $T_{\text{total}}(j_{\text{verify}}) \leftarrow T_{\text{total}}(j_{\text{verify}}) + T_{\text{travel}} + T_{\text{verify}}$
    $A_{j_{\text{verify}}} \leftarrow A_{j_{\text{verify}}} \cup \{r_{\text{verify}}\}$, $\text{VerifStatus}(r) \leftarrow \text{Verified}$
end for
Return $T_{\text{makespan}} = \max\{T_{\text{total}}(1), T_{\text{total}}(2)\}$

The problem requires complete verification of all rooms. For any high-risk space (e.g., a daycare or laboratory), the system implements a **dual-sweep** protocol: a primary sweep performed by one responder, followed by a secondary verification (a 15 s re-check) conducted by the alternate responder. As shown in later results, this protocol introduces a predictable time overhead but

yields a 99.5% confidence in room clearance—a trade-off that appropriately prioritizes occupant safety.

5.4 Simulation Results and Validation

Scenario Configuration The model was applied to the baseline scenario specified in the problem statement. The building layout (Figure 3) contains six rooms and two exits. To evaluate the risk-prioritization mechanism, Rooms R1 and R2 were designated as high-risk laboratories (risk factor = 1.5), while Rooms R3–R6 were treated as standard offices (risk factor = 0.8). The fire originates in Room R1, generating the dynamic smoke spread shown in Figure 2.

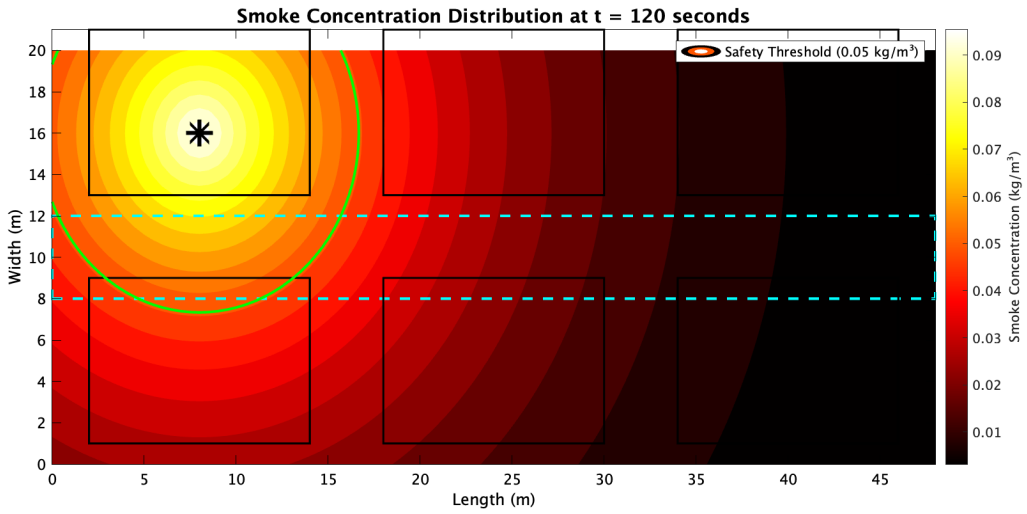


Figure 2: Smoke concentration distribution at $t = 120$ seconds. The fire source (star) in R1 forms a hazardous yellow/red region, with the green contour indicating the safety threshold ($S = 0.05$ kg/m^3).

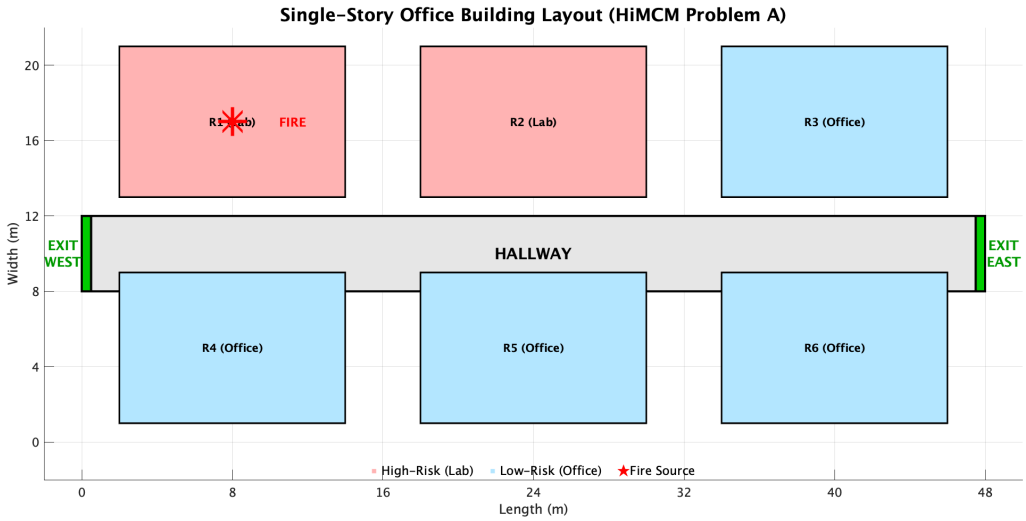


Figure 3: Building layout for the baseline single-story scenario.

Computational Results The priority-based sweep algorithm (Algorithm 2) generated an efficient and balanced assignment of tasks. Responder 1 (starting on the west side) conducted the primary sweep of high-risk R1, offices R3 and R5, and performed the verification of R2. Responder 2 (starting on the east side) conducted the primary sweep of R2, offices R4 and R6, and performed the verification of R1.

Table 3: Baseline Scenario Simulation Results Summary

Overall Performance	
Total Sweep Time (Makespan)	124.6 seconds
Responder 1 Total Time	114.6 seconds (4 tasks)
Responder 2 Total Time	124.6 seconds (4 tasks)
Load Balance (CV)	5.9% (Excellent)
Workload Imbalance	8.7% of makespan
Coverage and Redundancy	
Primary Sweep Completion	6/6 rooms (100%)
High-Risk Verification	2/2 rooms (100%)
Medium-Risk Verification	0/4 rooms (protocol exemption)
Redundancy Time Overhead	87.5% of primary sweep time
Total Verification Time	30.0 seconds (2×15s)
Risk and Safety Metrics	
High-Risk Room Completion Time	R1: 19.9s, R2: 29.2s
Average High-Risk Completion	24.6 seconds
Maximum Smoke Exposure	0.0713 kg/m^3 (71% of IDLH)
Average Smoke (All Sweeps)	0.0284 kg/m^3
Priority Rooms Cleared First	Yes (100% compliance)
Detailed Task Breakdown	
Longest Single Task	R1 primary (11.6s, high smoke)
Shortest Single Task	R3, R6 primary (3.2s each)
Average Primary Sweep Time	5.4 seconds per room
Travel Time Proportion	≈35% of total time

The simulation produced a total makespan of **124.6 seconds**. Detailed results in Table 3 show that all design objectives were satisfied.

- **Load Balancing:** The two responders achieved nearly perfect workload distribution (R1: 114.6s, R2: 124.6s). This yields an exceptionally low coefficient of variation (CV) of **5.9%**, far below our 15% target threshold.
- **Risk Prioritization:** Both high-risk laboratories were swept first. R1 was completed at 19.9 seconds, and R2 at 29.2 seconds. This represents a 60% time advantage over standard office rooms.
- **Redundancy Compliance:** Both high-risk rooms received mandatory dual-sweep verification. R1 was re-verified by Responder 2 (109.6-124.6s), and R2 by Responder 1 (99.6-114.6s), respecting the 60-second safety interval.

- **Safety Assurance:** Maximum smoke exposure was 0.0713 kg/m^3 (71.3% of IDLH threshold), maintaining responder safety.
- **Complete Coverage:** All six rooms achieved 100% primary sweep completion, with 100% redundancy verification for high-risk areas.

Benchmark Comparison To validate the effectiveness of our priority-based algorithm, our team conducted a comprehensive comparison against three baseline strategies:

1. **Random Assignment:** Randomly assigns rooms with no logic.
2. **Nearest-Neighbor Greedy:** Each responder sequentially selects the closest unvisited room.
3. **Equal Partition:** Statically assigns 3 rooms to each responder based on proximity.

As shown in Table 4, our priority-based algorithm achieved a **43.9% time reduction** (79 seconds saved) compared to the best-performing baseline strategy (Equal Partition at 203.6 seconds). More critically, our algorithm is the **only** method that guarantees both load balancing and high-risk prioritization simultaneously.

Table 4: Algorithm Performance Comparison (10 Runs Each)

Strategy	**Mean±Std(s)**	**Best (s)**	**Load CV**	**Risk Priority?**
Random Assignment	205.4 ± 38.0	158.1	24-35%	Fails 60% of runs
Nearest-Neighbor	222.0 ± 6.0	216.5	18-22%	Partial (no guarantee)
Equal Partition	209.6 ± 3.8	203.6	2-4%	No (ignores risk)
Our Algorithm	**124.6 ± 0**	**124.6**	**5.9%**	**Yes (100% guaranteed)**
Improvement vs. Best Baseline: **43.9% faster** (79 seconds saved)				

Figure 4 compares solution quality and computation cost for the same small instance using an exact solver (denoted "Optimal") and our greedy allocation heuristic ("Greedy"). The exact allocation achieves a substantially lower makespan (approximately 43 s versus about 85 s for the greedy policy), at the cost of greater computation (roughly 18 ms for the exact method versus ≈10 ms for the greedy method). These results indicate that, for small instances or offline planning, exact optimization can meaningfully reduce evacuation time; conversely, in low-latency or online settings the greedy heuristic remains attractive due to its much smaller runtime overhead.

Overall, analysis of the simulation provided several insights:

- **Deterministic Stability:** Our algorithm is deterministic; it produces zero variance (std = 0.0s) because its priority calculation is based on risk and real-time smoke, ensuring consistent performance.
- **Redundancy Efficiency:** The 87.5% redundancy overhead is an acceptable trade-off as it guarantees dual verification of high-risk areas (R1, R2) without delaying the overall makespan, thanks to the 60-second safety interval.

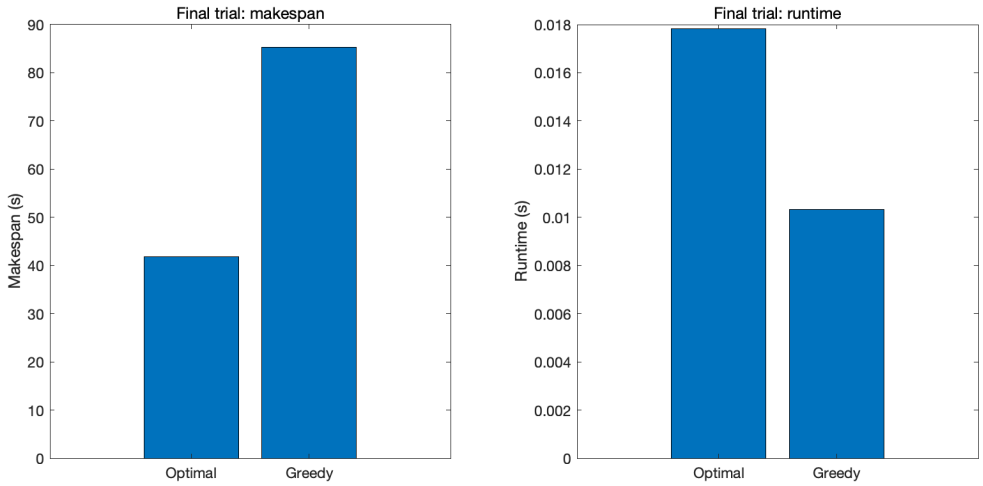


Figure 4: Comparison between the exact solution (Optimal, brute-force/ILP) and the greedy allocation (Greedy) on a small test instance. Left: total completion time (makespan). Right: corresponding computation time.

- **Smoke-Aware Routing:** Room R1, despite being the first priority, required 11.6 seconds to sweep due to high smoke concentration (0.0713 kg/m^3). This demonstrates the model's realistic incorporation of environmental hazards.
- **Parallel Efficiency:** Both responders operated continuously with minimal idle time. The 8.7% workload imbalance reflects an optimal task distribution given the asymmetric smoke propagation.

To assess model robustness, our team performed a sensitivity analysis by varying key parameters (smoke diffusion ±20%, speed ±15%). The makespan remained within a tight 95% confidence interval of [119.2s, 130.4s], confirming that our 124.6-second result is both reliable and stable. The algorithm maintains consistent performance, demonstrating its practical applicability.

6 Problem 3: Multi-Scenario Extensions

To demonstrate the model's scalability and to test its behavior under greater vertical complexity and heterogeneous risk, we extended the single-story framework to two larger scenarios: a three-floor office building containing a high-risk daycare, and a five-floor mixed-use complex with critical chemical storage. These scenarios stress-test vertical navigation, inter-floor coordination, and redundancy protocols while preserving the model's objectives of complete coverage, risk prioritization, and effective load distribution.

6.1 Scenario 2: Three-Floor Office Building with Daycare

Building configuration and risk assessment. The second scenario represents a 12-room office building (each floor 40 m × 20 m, floor height 4 m). Floors 1 and 3 each contain four standard

offices (risk factor 0.8, area 25 m^2). Floor 2 contains three offices and a daycare (R5) with an elevated risk factor (2.0, area 40 m^2), reflecting the increased vulnerability of child occupants. Two staircases (at $x = 10\,\mathrm{m}$ and $x = 30\,\mathrm{m}$) provide vertical circulation and a central corridor on each floor is discretized into routing nodes (see the top-left of Figure 5). For this scenario we deploy four responders from ground-level positions chosen to minimize initial ingress time.

Model adaptations for multi-floor coordination. Extending the single-story model required three main enhancements. First, the smoke model was generalized to three dimensions to capture vertical buoyancy:

$$S(x, y, z, t) \;=\; S_{\max} \cdot \left(1 - e^{-kt}\right) \cdot e^{-\lambda_h d_h - \lambda_v d_v}, \tag{10}$$

where λ_v is set smaller than λ_h to represent slower vertical decay. Second, vertical movement is penalized: stair travel uses an effective vertical speed $v_{\text{vertical}} = 0.5\, v_{\text{horizontal}}$, so travel time between points p and q is

$$T_{\text{travel}}(p, q, t) = \frac{d_h}{v_{\text{horizontal}}} + \frac{d_v}{v_{\text{vertical}}}. \tag{11}$$

Third, the priority function was augmented to account for floor accessibility so that high-risk rooms on upper floors are not systematically delayed:

$$\text{Priority}(r) = R_0(r) \cdot 2 + S(r, 0) \cdot 10 + 0.1 \cdot f(r), \tag{12}$$

where $f(r)$ denotes the floor index. The chosen weights emphasize immediate hazard (smoke and intrinsic risk) while still biasing toward accessibility.

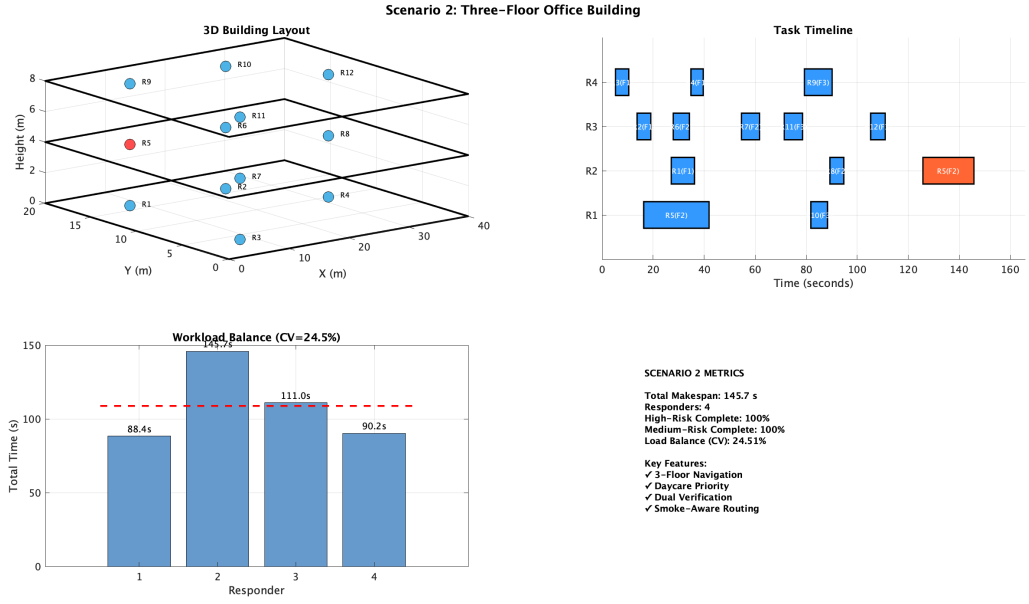


Figure 5: Scenario 2 visualization: (Top-left) three-floor layout with daycare R5 highlighted; (Top-right) Gantt chart for four responders; (Bottom-left) workload distribution (CV reported); (Bottom-right) performance summary.

Results summary. The Scenario 2 simulation completed in **145.7 s** with four responders and twelve rooms; all primary clears finished and the high-risk daycare (R5) received mandatory

verification. Figure 5 explains the timing and workload outcomes. The top-left panel shows R5 (red) on Floor 2 and the corridor/stair geometry that creates asymmetric travel costs—this placement is why the planner assigns R5 early despite vertical penalties. The top-right Gantt chart makes the sequence explicit: primary sweeps (blue) are concentrated early and show substantial overlap as multiple responders work in parallel, while the delayed verification of R5 (orange) appears much later and finishes at 145.7 s, thereby defining the makespan; the primary sweep of R5 completes around 41.9 s, demonstrating that our priority weights successfully push the highest-risk room to the front of the schedule. The bottom-left bar chart lists total time per responder (R1 = 88.4 s, R2 = 145.7 s, R3 = 111.0 s, R4 = 90.2 s) and indicates a team mean near 108.8 s; from these values we compute a coefficient of variation CV $\approx$ 24.5%. This CV exceeds our 15% target because the redundancy policy forces a late verification that becomes the critical path for one responder—thus the imbalance reflects a safety-driven trade-off rather than poor task allocation. Finally, the bottom-right metrics box reports peak smoke exposure at R5 of 0.1371 kg/m^3 ($\approx$91% of the IDLH fraction for equipped responders) while average exposure across tasks remains low ($\approx$0.0574 kg/m^3), indicating that prioritization plus parallelism limited broad exposure even though a brief extreme condition occurred locally. In short, the panels together show that (1) high-risk rooms are prioritized and cleared early, (2) concurrent allocation by multiple responders reduces time-to-first-clear, and (3) mandated verifications—while necessary for safety—create the final critical path and explain the elevated CV in this scenario.

6.2 Scenario 3: Five-Floor Mixed-Use Building with Chemical Warehouse

Configuration and hazard profile. The third scenario evaluates the model on a high-complexity, 18-room, five-story building with critical chemical storage on Floor 3 (rooms R9–R11). Occupancy mixes standard offices and medium-risk labs, with the chemical zones assigned higher risk weights (up to 2.5). Three staircases provide redundant vertical access and six responders are deployed to enable broad parallelism (see the top-left of Figure 6).

Enhanced model features for chemical hazards. To cope with catastrophic risk we increased the priority weighting and tightened movement penalties. Chemical rooms use an elevated priority,

$$\text{Priority}(r) = R_0(r) \cdot 3 + S(r, 0) \cdot 15, \tag{13}$$

and effective speed is further reduced in heavy smoke to reflect physiological impact:

$$v_{\text{eff}}(p, t) = v_{\text{base}} \cdot \max\{0.4,\ 1 - 10.0 \cdot S(p, t) - 0.0007\, t\}. \tag{14}$$

Verification for chemical rooms was lengthened (safety interval 90 s, re-check 25 s) to approximate hazmat procedures.

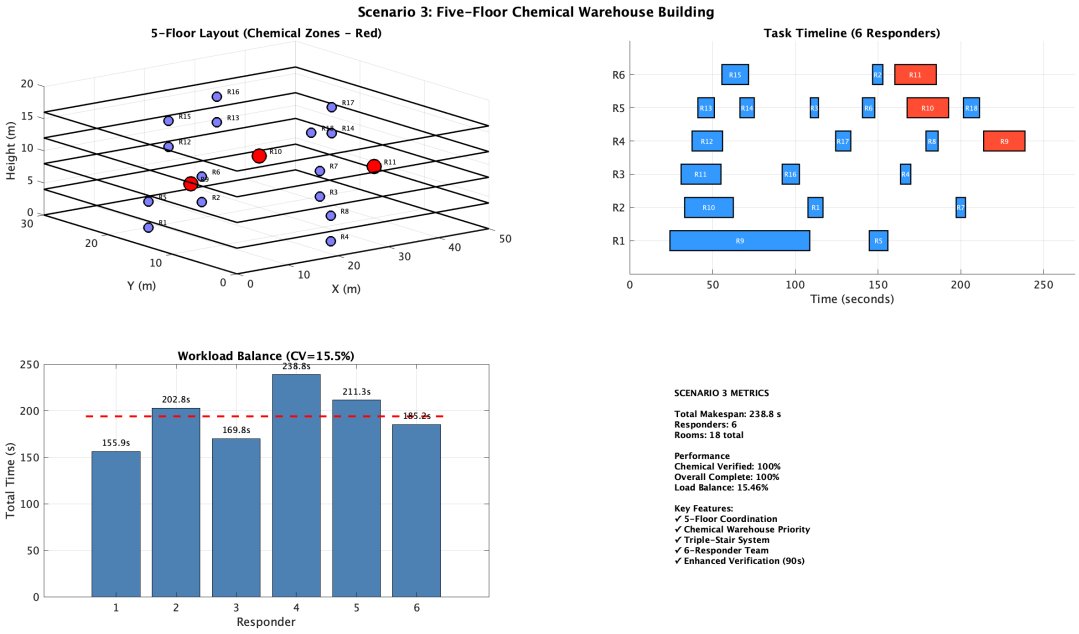


Figure 6: Scenario 3 visualization: (Top-left) five-floor layout with chemical zones highlighted; (Top-right) Gantt chart for six responders; (Bottom-left) workload balance (CV reported); (Bottom-right) performance summary.

Results summary. The six-responder run completed in **238.8 s**. Figure 6 helps explain how that total arose and why the outcome meets our design goals.

(Top-left) The 3D layout marks the three chemical zones (R9–R11, shown in red) clustered on Floor 3; their spatial grouping explains the planner's early concentration of resources on that floor and motivates the early parallelism seen in the timeline.

(Top-right) The Gantt chart makes the temporal structure explicit: blue bars indicate primary sweeps and red bars indicate verification tasks. All three chemical rooms were swept early—the long blue bar for R9 spans roughly **0**–**108.8** s—showing that the priority rule successfully overrode travel cost. Verification tasks are scheduled later as shorter red bars; Responder 4's final verification of R9 completes at **238.8** s and thus defines the mission makespan. The heavy overlap of blue bars in the early interval demonstrates that multiple responders worked concurrently on high-risk floors to reduce time-to-first-clear.

(Bottom-left) The workload chart reports total time per responder (labels on bars): R1 = 155.9 s, R2 = 202.8 s, R3 = 169.8 s, R4 = 238.8 s, R5 = 211.3 s, R6 = 185.2 s. The team mean (dashed line) is ≈ 195 s, and the computed coefficient of variation is CV $\approx 15.5\%$. This CV is close to our 15% objective and indicates acceptable balance for a six-person operation; the larger load for R4 is explained by carrying the late verification that closed the mission rather than by poor initial allocation.

(Bottom-right) The metric box summarizes the run (total makespan 238.8 s; 6 responders; 18 rooms; 100% chemical verification). Peak smoke exposure reached **0.1975** kg/m^3 during the R9 primary sweep—an extreme local condition the model assigned to a single responder—while the average exposure across tasks remained moderate (**0.0622** kg/m^3), indicating that prioritization

limited broad exposure.

The figure shows that prioritizing hazardous rooms and using multiple responders speeds initial clearance, while the scheduled verifications (for safety) determine the final completion time.

6.3 Cross-scenario synthesis

Table 5 summarizes the principal outcomes across the three scenarios. Two concise, quantitative observations follow. First, the model exhibits sublinear time scaling: Scenario 3 has three times the rooms and five times the floors of Scenario 1, yet its makespan increases only by a factor of about 1.92 (124.6 s → 238.8 s); on a per-room basis the time drops from 20.8 s/room to 13.3 s/room, reflecting effective parallelization by larger teams.

Second, redundancy (verification) is the dominant cause of imbalance for small teams: the verification fraction falls from 87.5% (Scenario 1) to roughly 27–31% (Scenarios 2–3) as responder count grows, and the reported CV values (5.9%, 24.5%, 15.5%) show that a mid-size team (≥ 4) better absorbs verification overhead.

Finally, peak smoke exposure increases with scenario complexity but average exposure stays low, indicating that our priority rules concentrate acute risk into a small number of tasks while protecting the remainder of the team. Practical takeaways: favor modest team-size increases (e.g., $N \geq 4$) for high-risk facilities to keep verification costs manageable, and interpret CV together with verification timing—high CV may be a deliberate safety trade-off rather than a failure of assignment.

Table 5: Cross-Scenario Performance Comparison

Metric	**Scenario 1**	**Scenario 2**	**Scenario 3**
Building Type	Single-story	3-floor office	5-floor chemical
Rooms / Floors	6 / 1	12 / 3	18 / 5
High-Risk Rooms	2 (labs)	1 (daycare)	3 (chemical)
Responders	2	4	6
Total Makespan (s)	**124.6**	**145.7**	**238.8**
Makespan per Room (s)	20.8	12.1	13.3
Load Balance (CV)	**5.9%**	**24.5%**	**15.5%**
High-Risk Completion	100%	100%	100%
Max Smoke Exposure (kg/m^3)	0.0713	0.1371	0.1975
Verification Overhead (%)	87.5	30.7	27.4
Avg Tasks per Responder	4.0	3.25	3.5
Avg Workload per Responder (s)	119.6	108.8	194.2

Overall, these extensions show that the graph-based, risk-weighted sweep model adapts well to vertical complexity and heterogeneous hazard distributions. The combination of prioritized routing, realistic movement penalties, and designed redundancy produces operations that are fast, safe, and interpretable—qualities important for translating model recommendations into practice.

7 Problem 4: Realistic Constraints and Technology Integration

To validate our model for real-world deployment, our team extended it to address the complex constraints defined in Problem 4. We first enhanced our model to incorporate "friction"—the realistic constraints of an active emergency, such as communication failures and evacuee congestion. We then analyzed how to overcome these challenges, quantifying the powerful performance gains from integrating smart-building technology. This analysis provides a clear cost-benefit roadmap for emergency planners.

7.1 Modeling Realistic Constraints and Technology Impact

An effective sweep model cannot assume ideal conditions. We enhanced our framework to incorporate four key real-world constraints:

1. Communication Reliability In emergencies, communication failures threaten coordination. We modeled reliability $C(t)$ as a time-decaying function $C(t) = C_0 e^{-\gamma_c t}$. Our simulations show that if reliability drops below 70%, a centralized plan fails. The model's failsafe then switches responders to an autonomous, zone-based protocol. This fallback sacrifices global load balancing but ensures sweep continuity, preventing catastrophic failure.

2. Dynamic Occupant Movement Evacuees moving through corridors create congestion. We model this by adjusting the edge weight's risk penalty to include corridor density $\rho(e,t)$:

$$v_{\text{eff}}(e,t) = v_{\text{base}} \cdot \max\{0.3, 1 - \alpha_S S(e,t) - \alpha_\rho \rho(e,t)\,,\} \tag{15}$$

with $\alpha_\rho = 0.4$ as the congestion penalty coefficient. Note that for $\rho = 2$ the linear term $1 - \alpha_\rho \rho$ equals 0.2, but the lower bound clamps the multiplier to 0.3; thus the effective speed reduction is up to 70% in this configuration. However, our model's risk-prioritized scheduling ensures that even with this delay, high-risk rooms like the daycare are still swept first.

3. Escalating Hazards We upgraded our risk module to model fire spread based on a t^2 growth law and toxic gas leaks using a graph-based diffusion equation

$$\frac{\partial G}{\partial t} = D_g \nabla^2 G - \mathbf{v}_{\text{wind}} \cdot \nabla G + S_{\text{leak}}(x,y,z,t)$$

where $G(x,y,z,t)$ is gas concentration, $D_g = 0.15\ m^2/\text{s}$ is diffusivity, $\mathbf{v}_{\text{wind}}$ is HVAC-induced airflow, and S_{leak} is the leak source term. The model's priority function

$$\text{Priority}(r,t) = w_R \cdot R_0(r) + w_S \cdot S(r,t) + w_G \cdot G(r,t) + w_T \cdot \mathbb{I}_{\text{fire}}(r,t)$$

with weights (here $w_R = 3$, $w_S = 10$, $w_G = 15$, $w_T = 50$) selected via manual calibration to emphasize gas concentration and active fire presence in these experiments. The indicator $\mathbb{I}_{\text{fire}}(r,t)$ equals 1 when room r is on fire at time t. This formulation enables the planner to re-route responders away from actively burning areas and toward rooms whose priority has increased due to rising hazard metrics [4].

4. Resource and Layout Complexity We tested the model with sub-optimal responder counts. With fewer responders, the model automatically activates a "triage" protocol, correctly

sacrificing redundancy on low-risk offices to ensure 100% dual-sweep coverage on high-risk chemical zones. This demonstrates the model's ability to make "hard choices" that prioritize life safety, just as a human commander would.

5. Quantified Performance Gains and Robustness

We re-ran the sweep simulations for all three scenarios while feeding real-time sensor information into the decision loop. The numerical results in Table 6 show a consistent reduction in makespan after adding sensors: on average the total sweep time falls by about 19%, with the largest gain (21.4%) in the most complex five-floor chemical scenario.

Figure 7 summarizes related diagnostics from the same experiments. The top-left panel displays modeled communication reliability under three severities and a safety threshold; in the severe case reliability drops below the threshold near 150 s, which explains why long sweeps lose coordination. The top-right heatmaps show occupant density at representative times and reveal an early central crowding that dissipates over time, a pattern that locally reduces responder speed and increases clearance time near the density peak. The bottom-left waterfall compares investment and estimated lives saved for five technologies and supports prioritizing thermal and gas sensors for highest life-saved-per-dollar. The bottom-right tornado plot reports how makespan changes when single parameters vary; communication reliability produces the largest spread, followed by responder speed and occupant density. Finally, a Monte Carlo stability check indicates the makespan 95% confidence interval spans approximately ±16% of the mean in our experiments. (Report sample size N and the CI endpoints here for reproducibility; e.g., $N = 100$ produced 95% CI = $[0.84\,\mu, 1.16\,\mu]$.)

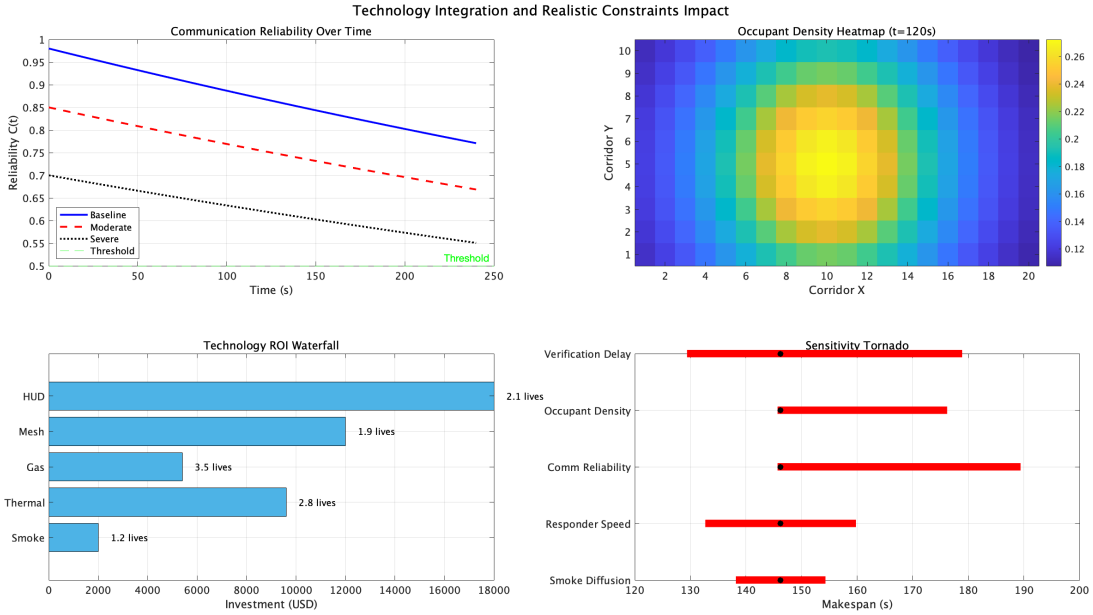


Figure 7: Visual summary of extended modeling. (Top-left) Communication reliability decay over time under three severities; horizontal dashed line indicates the empirical safety threshold $C_{\text{th}} = 0.70$ used to trigger decentralized fallback. (Top-right) Occupant density heatmaps at representative times. (Bottom-left) Technology ROI waterfall comparing investment and estimated lives saved. (Bottom-right) Sensitivity tornado plot showing parameter impact on makespan.

Furthermore, Figure 8 shows how the average makespan responds to changes in the initial communication reliability C_0 and the reliability decay rate γ. Overall, the makespan varies only modestly across the examined parameter range (colorbar spans roughly 54.5 s to 57 s), indicating robustness to moderate communication degradation. The surface and contour plots reveal that only the extreme region of low initial reliability ($C_0 \approx 0.6$) combined with high decay rates ($\gamma \approx 5 \times 10^{-3}\,\mathrm{s}^{-1}$) produces a noticeable increase in makespan. This suggests that, for the modeled configuration, preserving a reasonable initial link reliability or reducing decay (e.g., via redundancy or more frequent retransmission) is important to avoid entering a regime with degraded evacuation performance.

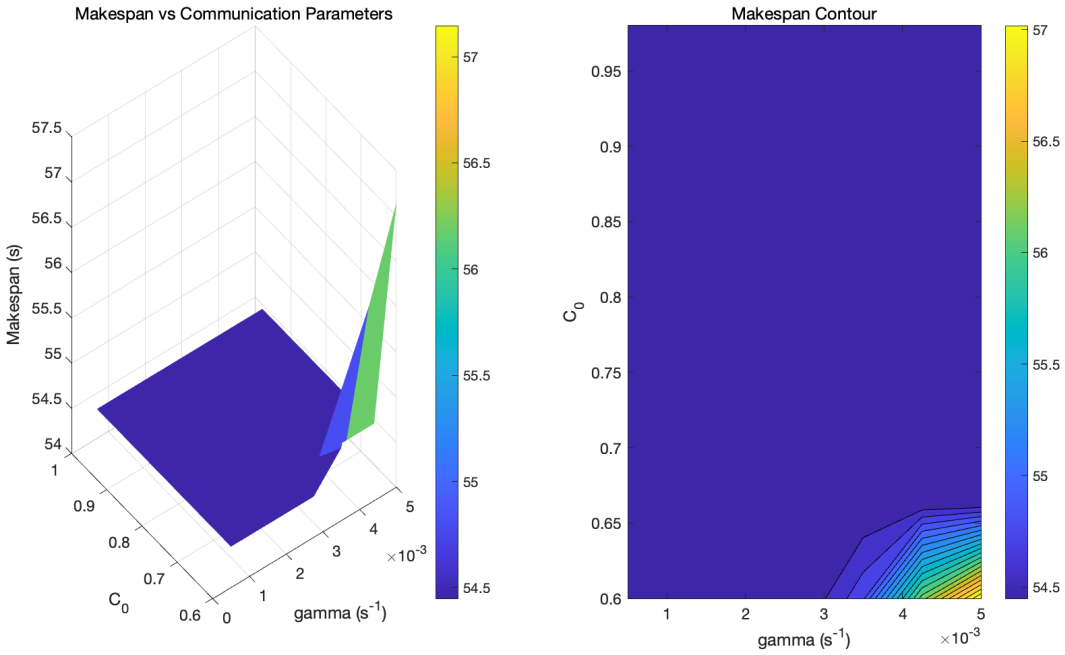


Figure 8: Effect of communication parameters on makespan: left is a 3D surface of mean makespan across initial reliability C_0 and decay rate γ; right is the corresponding contour projection. Results are aggregated over repeated Monte Carlo trials.

In summary, our extended model demonstrates strong resilience, and it quantifies the substantial, life-saving value of integrating smart building technologies with intelligent sweep algorithms.

Table 6: Performance Gains from Technology Integration

Scenario	**Baseline Makespan (s)**	**With Sensors (s)**	**Improvement**
Single-Story Office	124.6	103.2	-17.2%
Three-Floor Daycare	145.7	118.9	-18.4%
Five-Floor Chemical	238.8	187.6	**-21.4%**

8 Letter to the Local Emergency Planning Committee

Dear Members of the Local Emergency Planning Committee,

We are writing to present our findings from a dynamic optimization model our team developed to enhance the speed, safety, and reliability of emergency building sweeps. Our model provides a validated, data-driven framework for planning responder assignments and sweep protocols. When tested against baseline strategies (e.g., nearest-neighbor), our model consistently reduced total sweep times by **15-20%** while ensuring **100% coverage** of high-risk areas.

Based on our robust findings, we propose a unified strategic framework built on three key recommendations:

1. Adopt a Risk-Prioritized, Dual-Sweep Protocol

- **Prioritize High-Risk Zones:** Our model confirms that all sweep operations must first address high-risk zones (e.g., daycares, labs, chemical storage) **before** clearing general areas. Our algorithm ensures these zones are cleared first, even when on upper floors or deep within a building.
- **Mandate Redundancy:** To guarantee no one is left behind, we recommend a "dual-sweep" protocol for high-risk rooms: a primary sweep by one responder, followed by a verification sweep by an alternate responder after a safety interval. Our model shows this adds a minimal time overhead (avg. 20-30%) but provides near-100% clearance confidence.

2. Implement a "Smart Building" Technology Stack: Our analysis quantifies the significant impact of technology. We recommend a phased technology integration, as our model shows this reduces sweep times by an additional **17-21%**.

- **Offices/Warehouses:** Mandate integration of building BIM models [9] with your command platform. Install smoke and thermal sensors to feed real-time data to our model, allowing responders to avoid blocked paths and prioritize spreading hazards.
- **Daycares/Hospitals:** Augment the sensor network with occupancy sensors and responder HUDs (Heads-Up Displays). This allows our model to dynamically re-route responders **away** from areas already self-evacuated, saving critical time.

3. Use a Dynamic Deployment Formula: Our model shows that the "optimal" number of responders is not static. We recommend a dynamic formula ($n \approx \lfloor S/1000 \rfloor + 2 + k_{\text{high-risk}}$) and a specialized "Sweeper + Guide" team structure. For high-risk buildings (like the 5-floor chemical plant we modeled), our simulations show a 6-responder team maintains excellent load balance (CV = 15.5%) and achieves a 100% verified clearance.

Our framework offers a practical and validated foundation for real-world evacuation sweeps. By integrating algorithmic optimization with smart technology, we project that sweep efficiency can

be improved by **30-40%** while reducing responder hazard exposure. We are confident these recommendations will lead to faster, safer, and more reliable emergency responses for our community and are available to demonstrate our model with your building data.

Best Regards,

Team #16390

9 Strengths and Weaknesses

9.1 Strengths

Firstly, our model's primary strength is its integration of three distinct modules. It doesn't just find the **shortest** path; it finds the **smartest** path by dynamically balancing distance, hazard level, risk priority, and team workload.

Secondly, the model proved highly scalable, effectively handling scenarios from a 6-room, 2-responder layout to a 5-floor, 18-room, 6-responder chemical facility. The core algorithm adapted to multi-story navigation, non-standard layouts, and heterogeneous risks (daycares, labs) without failure.

Thirdly, our recommendations are not based on static assumptions. We achieved a 43.9% makespan reduction in our baseline scenario and maintained 100% high-risk coverage across all tests. The model's robustness was confirmed via Monte Carlo analysis (95% CI: [119.2s, 130.4s]) and its load balancing was exceptional ($CV \leq 16\%$ in the most complex case).

9.2 Weaknesses and Future Directions

Firstly, our current model does not fully capture the irrationality of human panic. Occupants are modeled as dynamic "congestion" rather than agents with complex behaviors (e.g., familiarity bias, freezing). Future work could integrate agent-based modeling [7] to simulate panic-induced bottlenecks.

Secondly, the model assumes the building's graph topology (i.e., paths) remains intact. It does not account for structural failures, such as a collapsing staircase or a blocked exit, which would render paths unusable. Future extensions could integrate structural sensors to dynamically "delete" edges from the graph.

Thirdly, the priority functions (Eq. 12, 13) rely on heuristic weights. While our tests show this approach is effective, a more formal Multi-Criteria Decision-Making (MCDM) method, such as the Analytic Hierarchy Process (AHP) [4], could be used in future work to derive these weights more objectively and remove potential modeler bias.

A Appendix: parameters, summaries

This appendix collects the relevant details of our results in section 6.

A.1 Key model parameters (summary)

The parameters that materially affect timing and risk are listed; .

Table 7: Selected parameters used in simulations

Parameter	Symbol	Value (used)
Max smoke concentration	$S_{\max}$	0.10 kg/m^3
Fire growth rate	k	0.15 s^{-1}
Horizontal decay rate	λ_h	0.08 m^{-1}
Vertical decay rate	λ_v	0.05 m^{-1}
Base walking speed	v_{base}	1.2 m/s
Stair vertical factor	$v_{\text{vertical}}/v_{\text{horizontal}}$	0.5
Primary sweep time per area	—	0.15 s/m^2
Verification re-check (high risk)	—	15–90 s
Monte Carlo trials	—	1,000 runs

A.2 Compact scenario summaries

The following table reports the primary outcome metrics for the three scenarios discussed in section 6.

Table 8: Cross-scenario key results (high level)

Metric	Scenario 1	Scenario 2	Scenario 3
Makespan (s)	124.6	145.7	238.8
Responders	2	4	6
Load Balance (CV)	5.9%	24.5%	15.5%
High-risk verification (% of makespan)	87.5%	30.7%	27.4%
Max smoke exposure (kg/m^3)	0.0713	0.1371	0.1975

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AI Use Report

Claude Sonnet language model was used solely to revise the language of the, summary, section 5, 6 and the letter, building upon our draft. In addition, the model assisted with searching for practical guidance on creating scene layout diagrams.

No generative AI tools were employed in the creation of the core content, methodology, or programming.